

Networking & Internet Fundamentals

A Quick, Non-Technical Reference Sheet

1 Why This Matters

You don't need to be a network engineer to understand how the internet works. This sheet covers the terms you'll keep bumping into – IP, MAC, HTTP/HTTPS, DNS, SSH, ports – explained simply, with just enough detail to sound informed in interviews or team conversations.

2 IP Address

An **IP (Internet Protocol) address** is like a home address for a device on a network. It tells other devices where to send data.

IPv4: e.g. 192.168.1.10 – four numbers (0–255) separated by dots.

IPv6: e.g. 2001:0db8:85a3::8a2e:0370:7334 – newer, longer format because IPv4 addresses ran out.

- **Public IP:** your address on the wider internet (assigned by your ISP).
- **Private IP:** your address inside your home/office network (e.g. 192.168.x.x).
- **Static IP:** never changes. **Dynamic IP:** changes periodically.

3 MAC Address

A **MAC (Media Access Control) address** is a unique ID burned into a device's network hardware (like a serial number for your Wi-Fi/Ethernet card).

Format: 00:1A:2B:3C:4D:5E – six pairs of hex digits.

Unlike an IP address, a MAC address doesn't change based on the network you join.

IP vs MAC (quick comparison):

Feature	IP Address	MAC Address
Assigned by	Network/ISP	Manufacturer
Can change	Yes (dynamic)	No (fixed)
Used for	Routing across networks	Identifying device on local network
Example	192.168.1.5	00:1A:2B:3C:4D:5E

4 HTTP vs HTTPS

HTTP (HyperText Transfer Protocol) is how browsers and servers talk to each other to load web pages.

HTTPS is the same thing, but **Secure** – data is encrypted using **SSL/TLS**, so no one can snoop on it in transit.

HTTP: `http://example.com` – unencrypted, data sent in plain text.

HTTPS: `https://example.com` – encrypted, shows a padlock icon in the browser.

Why it matters: passwords, card details, and personal data should only ever be sent over HTTPS.

5 DNS – Domain Name System

Humans remember names, computers use numbers. **DNS** translates a website name into an IP address.

You type: `www.google.com`

DNS resolves it to: something like `142.250.190.14`

Your browser then connects to that IP.

Think of DNS as the internet's phonebook.

6 Ports

A **port** is like a specific “door” on a device that a particular service uses to communicate. The IP address gets data to the right *house*; the port gets it to the right *room*.

Port	Common Use
20/21	FTP (file transfer)
22	SSH (secure remote login)
25	SMTP (sending email)
53	DNS
80	HTTP
443	HTTPS

7 SSH – Secure Shell

SSH lets you securely log into and control another computer remotely over a network, with everything encrypted.

Example command: `ssh username@192.168.1.10`

Commonly used by developers to access servers, and in game dev to manage build/deploy servers.

8 SSL / TLS

SSL (Secure Sockets Layer) and its modern successor **TLS (Transport Layer Security)** are protocols that encrypt data sent between a browser and a server – this is what makes HTTPS “secure.”

- Protects against eavesdropping and tampering.
- Verified using digital **certificates** issued by trusted authorities.
- That padlock icon in your browser = valid SSL/TLS certificate.

9 Other Terms Worth Knowing

Term	Meaning
Router	Directs traffic between your network and the internet
Firewall	Filters incoming/outgoing traffic for security
VPN	Creates an encrypted tunnel to browse privately/securely
Bandwidth	Max amount of data that can move through a connection
Latency	Delay between sending and receiving data (important in online games!)
Packet	A small chunk of data sent over a network
TCP	Reliable, ordered data delivery (used for most web traffic)
UDP	Faster, no guarantee of delivery – used in real-time games/voice chat

10 Quick Recap

IP = device's address on a network
MAC = permanent hardware ID of a device
DNS = translates website names to IP addresses
HTTP/HTTPS = how web pages load (S = secure/encrypted)
Port = specific channel for a type of traffic
SSH = secure remote access to another machine
SSL/TLS = encryption behind HTTPS